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Postgraduate Diploma Data Analytics

Programming for Data Analytics

ST10545405

POE PART 2

Dreaded Disease Benefit Triage: Client Report

Dataset: cancer_patient_data.csv

1. Executive Summary

The medical scheme brief is to expedite the application of cancer related dreaded disease benefit to customers who require them. The dataset provided allows for a proof-of-concept classification of recorded levels of risk (Low, Medium and High). No cancer diagnosis has been confirmed, no policy terms have been confirmed, no documentation of claims has been received, benefits were not confirmed or determined. Case triage and prioritisation can thus be demonstrated, but benefit entitlement cannot be determined by the model. The final analysis is done using a group aware train/test split to avoid having the same predictive profile in both training and test sets, as well as feature selection derived from Random Forest followed by a logistic regression benchmark, and baseline and refined versions of the Random Forest classifiers. For the 197-row holdout, the chosen Random Forest model was able to attain 94.9% accuracy, 96.2% balanced accuracy, and 0.958 macro-F1 score. All Low cases were correctly identified, 78 cases were classified as High and 10 cases as Medium. The results are academically impressive but caution is needed due to the small number of unique predictor profiles (152) of 1,000 rows.

2. Justification of Dataset Choice

2.1 Quality

The dataset is suitable for a basic classification exercise because it is structured, labelled and complete. The file contains 1,000 records and 25 columns: one patient identifier, 23 candidate predictors and the target Level. The quality review found no missing values and no exact duplicate rows. Numeric coding reduces preprocessing complexity, and all three target classes are represented: Low 303, Medium 332 and High 365.

A significant quality constraint is that the 1,000 rows contain only 152 unique predictor profiles after excluding Patient Id and Level. Different IDs frequently share identical values. A normal random row split could therefore place the same profile in training and testing and inflate performance. The analysis addresses this by grouping identical profiles and placing each profile wholly in either training or testing.

2.2 Suitability for the brief

The dataset is structured, labelled, complete and is suitable for a basic classification exercise. The file has 1000 records, and 25 columns: one patient identifier, 23 candidate predictors and one target value: Level. The quality check revealed no missing data and no exact duplicate records. Numerical coding helps minimize the complexity of the pre-processing and represents all three target classes: Low 303, Medium 332 and High 365. One of the limitations of the quality is 1,000 rows with only 152 unique predictor profiles (excluding Patient Id and Level). IDs can often have the same values. This would lead to over-optimism about performance if a normal random row split were used, for example. To overcome this, analysis will be done by aggregating the same profile and keeping each profile fully in the training set or testing set.

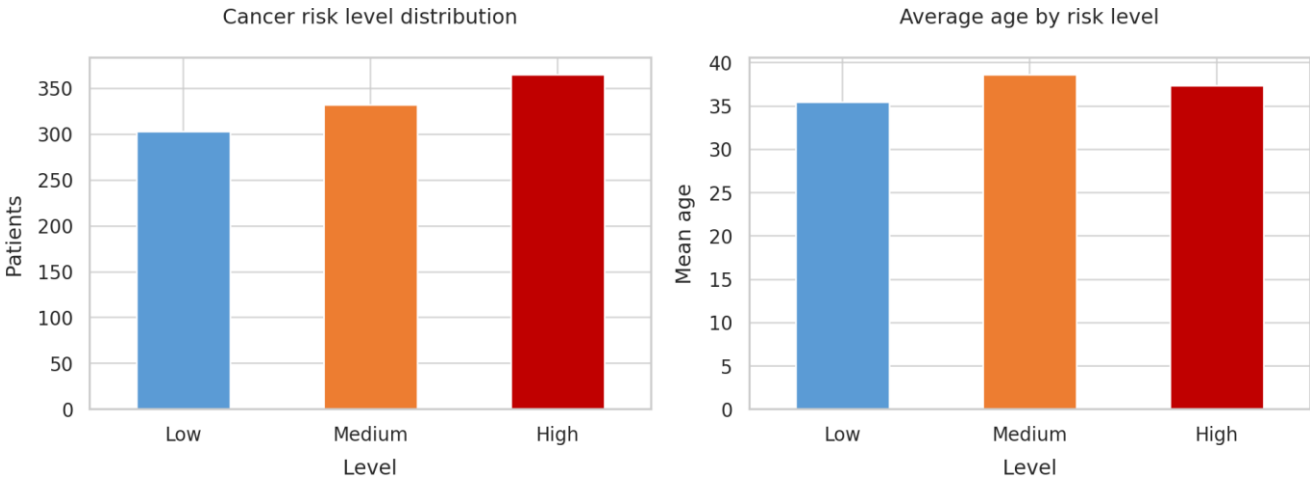


Figure 1. Class distribution and age profile in the supplied dataset.

3. Justification of Algorithm Choice

The dataset is applicable to the requested basic cancer analysis, as it contains a labelled three-class outcome and predictors of age, environmental exposures, lifestyle, medical history and symptoms. This facilitates exploratory analysis, feature selection, multiclass model training and evaluation. High, Medium and Low may be used as a prioritisation scale for directing cases for review as a prototype. The data set is not adequate to apply benefits. Level is not a confirmed diagnosis, clinical stage, policy eligibility indicator, approved claim and/or payment outcome. Provenance, methods of data collection, definitions of the scores and clinical validation of the results are not provided either. The correct answer is that the dataset is suitable for an academic proof of concept but for a production model validated scheme claims and policy and clinical outcome data must be used

The main classifiers selected for the analysis were Random Forest, as the objective is to classify each case into one of three classes and all the predictors are numeric scores that can have non-linear relationships and interactions. Tree ensembles can reproduce such patterns without requiring the predictors to be linear or normally distributed. Random Forest also has feature-importance values that provide information about the fitted model, is reasonably robust to correlated inputs and has class weights support. After training, predictions can be made quickly, fitting a triage process that will route cases quickly. As a clear reference, the multinomial logistic regression was kept. This comparison will determine if the gains in flexibility of random forest are worth the effort. At macro-F1, random forest was chosen, however the feature importances found here are predictive and not causal and cannot be used for clinical or policy interpretation.

.4. Analysis Plan

Stage	Planned activity	Success evidence
EDA and cleaning	Check shape, types, missing data, duplicates, ranges, target balance and class patterns.	Documented clean table and quality risks.
Leakage control	Remove Patient Id and group identical profiles before splitting.	No profile shared across train and test.
Feature selection	Fit Random Forest selection on training data only; retain features at or above median importance.	Reduced, reproducible feature set.
Model training	Train logistic benchmark, baseline forest and refined forest with balanced class weights.	Reproducible fitted models.
Evaluation	Compare accuracy, balanced accuracy, macro-F1, class metrics and confusion matrices.	Strong performance without hiding High-class misses.
Retraining	Increase trees and regularise leaves/max-features; compare with baseline.	Retain changes only if results are maintained or improved.
Reporting	Explain cleaning, EDA, selection, training, results, limitations and safeguards.	Client-readable evidence linked to notebook.

5. Exploratory Data Analysis and Cleaning

The whitespace for column names was stripped. Patient Id was removed due to the fact that an Id should not affect classification. There was no imputation required as all the fields were complete. There was a reasonably even distribution of classes for multiclass training. Class-level means indicated clear distinctions between a number of exposure, lifestyle and symptom variables, justifying the selection of a supervised classification. The patterns are correlations in the provided

labels and do not prove causality.

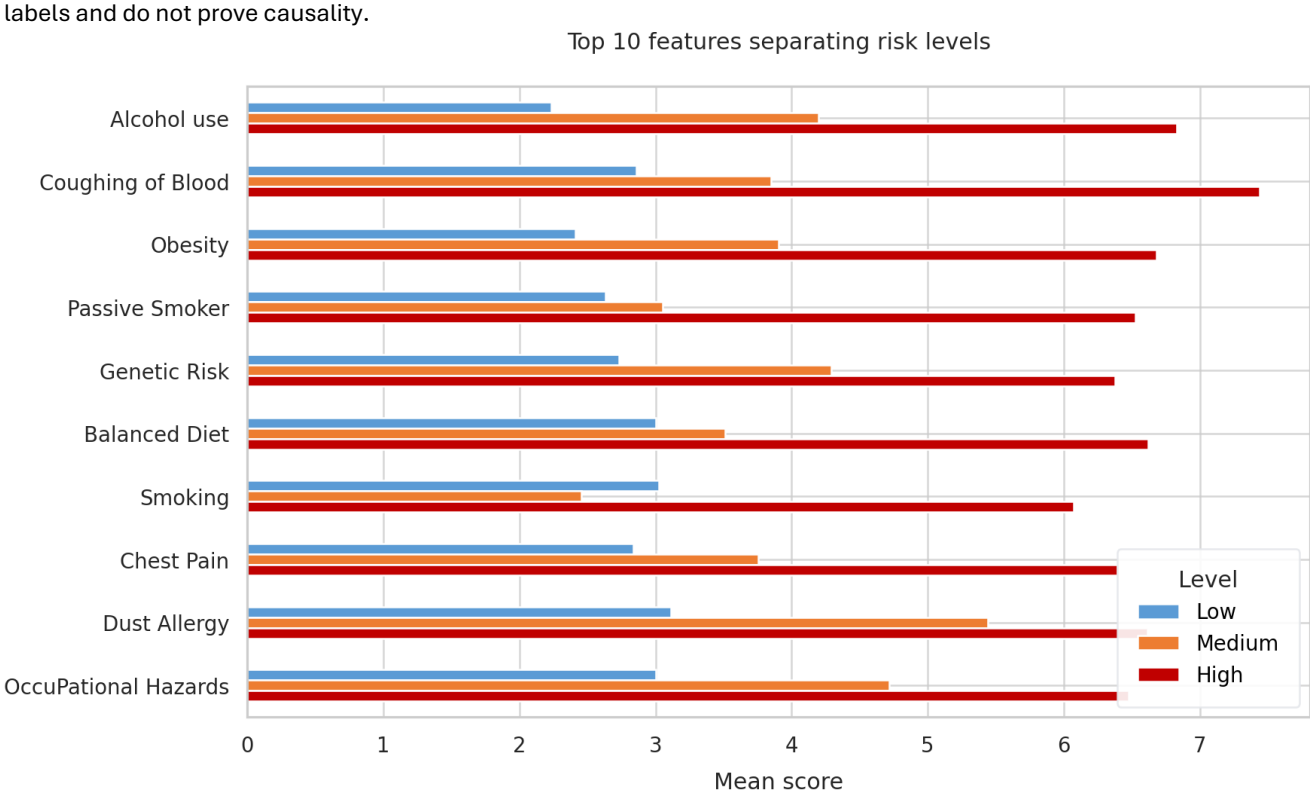


Figure 2. Variables with the largest mean-score separation across recorded levels.

6. Feature Selection

Feature selection was fitted on the training set only. A balanced 300-tree Random Forest estimated predictor importance and SelectFromModel retained features at or above the median importance. This embedded method fits the non-linear primary algorithm better than p-value elimination, which is tied to parametric regression assumptions.

Selected predictors: Air Pollution, Alcohol use, OccuPational Hazards, Balanced Diet, Obesity, Smoking, Passive Smoker, Coughing of Blood, Fatigue, Shortness of Breath, Wheezing, Swallowing Difficulty.

7. Model Training and Retraining

The profile-grouped holdout had 803 training rows and 197 testing rows; there were no common predictor profiles. All models were fitted using only 12 of the features and random_state=42. The logistic benchmark included standardisation and balanced class weights. 100 trees were used in the baseline forest. The refined forest used 500 trees, min_samples_leaf=2, max_features=sqrt and balanced class weights. More trees will give more stability and a minimum leaf size of 2 will give a little regularisation. The refined model returned the same result as the original model with no new complexity added to the model as an unsupported gain

8. Model Evaluation and Interpretation

Model	Accuracy	Balanced accuracy	Macro-F1
Logistic regression benchmark	0.954	0.919	0.934
Random Forest baseline	0.949	0.962	0.958
Random Forest refined	0.949	0.962	0.958

Accuracy is the overall percentage accuracy. Balanced accuracy is the average of the recall for each class and thus does not let the largest class dominate. Macro-F1 is the main comparison metric that ensures equal weight is assigned to the Low, Medium and High, and balances precision and recall. High class recall is also significant in case a case is missed, it could put the case on hold for review. The confusion matrix is used to make the direction of errors apparent

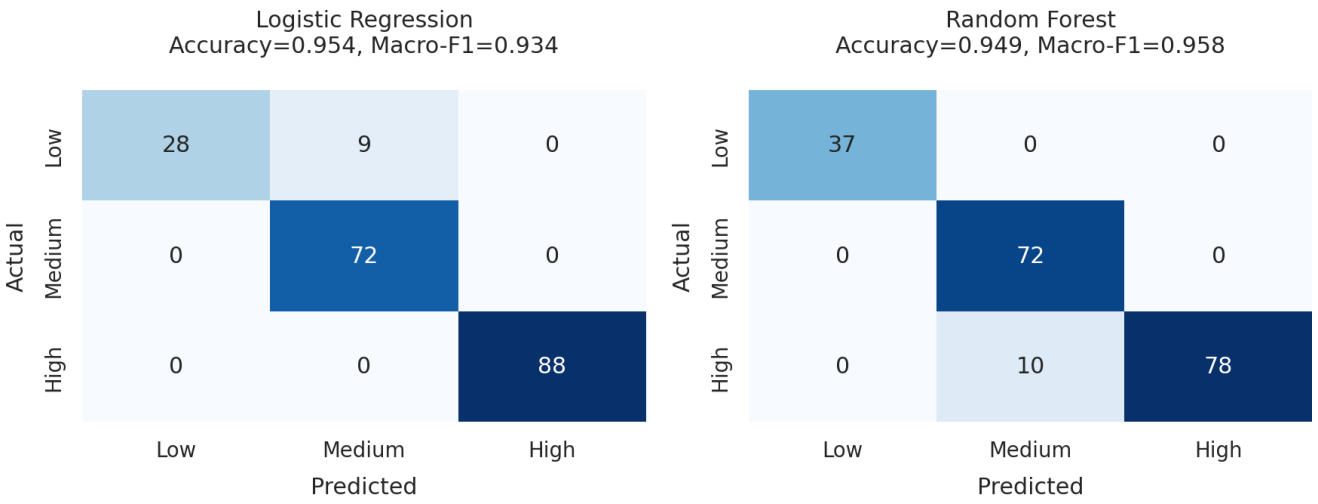


Figure 3. Holdout confusion matrices for the earlier benchmark comparison; the revised notebook provides benchmark, baseline and refined matrices using the selected variables.

. The achieved accuracy was 187 out of 197 rows in the holdout set for the selected Random Forest. The percentage of low, medium and high recall was 100%, 100% and 88.6% respectively. Medium was forecast for ten High rows. If a workflow is a benefit-triage, those errors are problematic, since they may reduce the priority of potentially important cases. Therefore, conservative probability thresholds and human review should be used instead of direct automation to operate an operational system..

9. Effectiveness, Limitations and Client Recommendation

The model works well as an academic proof-of-concept to train multiclass classification and can be used as a proof-of-concept prototype for prioritising cases for further investigation. It is easily reproducible, leakage from duplicate profiles has been managed, and its performance has been measured using both class-specific and overall performance metrics. The target Level is not a cancer or dreaded disease benefit decision. No data source information, sampling, coding definitions and clinical validation are provided. There are 152 unique predictor profiles in 1,000 rows. Analysis doesn't provide probability calibration, fairness and temporal stability or external validity. Feature importance is not to be interpreted as medical causation! The model shall not be allowed to approve or deny benefits, adjust premiums or to diagnose disease or to replace clinical and policy review. The scheme should clarify a validated target, e.g. adjudicated cancer benefit eligibility or claim outcome, prior to production use and train on its own de-identified policy, claim, diagnosis, staging, treatment, documentation and decision data. A final workflow is necessary that incorporates data-protection controls, probability calibration, validation of all recommendations by independent input, and monitoring and human review of all recommendations.

10. Reproducibility and Submission

The plan, commented code, EDA, feature selection, training, retraining and evaluation is available in the companion notebook. Does not rely on a computer-specific path, but instead relies on a relative-path search. On GitHub, save the notebook and the cancer_patient_data.csv to the same folder in the repository and install pandas and numpy in addition to the other packages listed above, then restart the kernel and execute all cells.